Dealing with Outliers

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When should we remove outliers?

- Often, when dealing with outliers, we face tough questions about whether to include those data or not.

- These decisions are highly contextual but in general, **it is not appropriate to drop a data point simply because it is an outlier**.

  - Rare events and/or exceptional individuals happen.

When should we remove outliers?

1. If it is obvious that an outlier is due to an error or comes from a different population, then it is safe to remove, but this removal should still be reported.

- E.g., let's say we find an outlying data point, we can remove it…
  - If it is a biologically impossible value (e.g., a human VO2max = 185).
  - If it came from an unusual event (e.g., a fire alarm went off during that data collection).
  - If it came from a fundamentally different population (e.g., your sample is intended to be PD patients off medication, but in retrospect you find one of them was on medication).

When should we remove outliers?

2. If the outlier does not change our result (magnitude/significance) but does affect our assumptions, you can drop the outlier, but its removal should be reported.

   - E.g., an outlying data-point creates positive skew in the residuals, leading to a violation of normality and homoscedasticity. Critically though, the inclusion of this data-point does not **substantially*** alter the magnitude or significance of the results.

* Note that substantial is a contextual judgment.

When should we remove outliers?

3. If an outlier affects our results and our assumptions, then the situation gets more complicated:
   - Usually, this will be captured by a large Cooks' Distance.

   A. If an outlier is driving a result/association, then that outlier should be removed and results should reported without the outlier included.
      - Relationships that disappear when an outlier is removed are not robust.

   B. If an outlier affects a result/association, then it might be reasonable to run the analysis with and without the influential data points.
      - Ideally, there should also be a compelling (non-statistical reason) for why those data points might be different.
      - All analyses should be reported.
      - Any differences in the analyses need to be emphasized.

Retaining outliers:

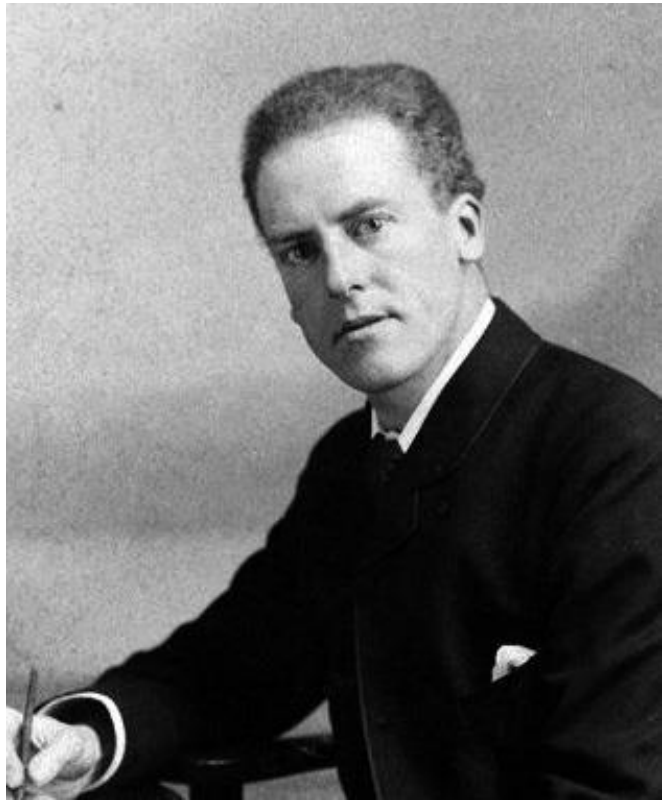
- Erring on the side of caution then, we often want to include all of the data, but reduce any disproportionate influence a single data-point has on the model.

- How can we retain data while meeting all of our assumptions?
  - Do we have the right statistical model?
    - (Choose an approach with different assumptions.)
  - Is a transformation of the outcome appropriate?
    - (Bend our data to fit our assumptions.)

- ***But if our result is being driven by anomalous data/only exists when our assumptions are violated, then it isn't really much of a finding and the removal of outliers is warranted***.

“Put your [data] on the table!”

- If you are confused or uncertain on what to do in a certain situation, **that is okay**!
- In research, we will often encounter ambiguity.
- In those situations we should be transparent, reporting all analyses and as much of the data as possible.
  - This allows readers/reviewers to make informed judgments and recommendations.



[Karl Pearson; wikicommons]